

Short Question 2013

Write C++ program Code Segment for the following:

(a) To calculate equivalent of resistance in series?

Program:

```
#include<iostream>
using namespace std;
main()
{
    float R,n,Req=0;
    cout<<"Enter the Number of Resistance In Series :\n";
    cin>>n;
    cout<<"Enter the value of Resistance:\n";
    cin>>R;
    for (int i=0;i<n;i++)
    {
        Req=Req+R;
    }
    cout<<"The equivalent resistance in series combination of same resistance =:\n"<<Req;
    return 0;
}
```

(b) To print “x” against “f(X)” for the equation

$$f(x)=3x^3-4x^2+2x+4$$

Program: (Simple)

```
#include<iostream>
#include<math.h>
using namespace std;
main()
{
    float x;
    float fx;
    cout<<"enter the value of x:\n";
    cin>>x;
    fx=3*pow(x,3)-4*pow(x,2)+2*x+4;
    cout<<"Value of x="<<x<<" & corresponding function="<<fx;
    return 0;
}
```

Program: (using array)

```
#include<iostream>
```

```

#include<math.h>
using namespace std;
main()
{
    int i,j,s;
    cout<<"enter the size of array : \n";
    cin>>s;
    float x[s],y[s];
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    cout<<"Values of X is="<<"\t\t"<<"and f(x)="<<endl;
    for (j=0;j<s;j++)
    {
        y[j]= 3*(pow(x[j],3))-4*(pow(x[j],2))+2*x[j]+4;
        cout<<"\t"<<x[j]<<"\t\t"<<y[j]<<endl;
    }
    return 0;
}

```

(c) To calculate the factorial of a number read from the user ?

Program:

```

#include<iostream>
using namespace std;
main()
{
    unsigned long fact=1;
    int i,n;
    cout<<"Enter the number:\n";
    cin>>n; //any number whose factorial you want to find
    for(i=1;i<=n;i++)
    {
        fact=fact*i;
    }
    cout<<"factorial of number is="<<fact;
    return 0;
}

```

(d) To convert temperature from Celsius to kelvin scale using a function ?

Program:

```

#include<iostream>
using namespace std;
float kelvin (int c)
{
    float k;
    k=c+273.16;
    return k;
}

main()
{
    float c;

    cout<<"enter the celcius temperature:\n";

    cin>>c;
}

```

```

cout<<"Temperaturein Kelvin:="<<kelvin(c);

return 0;

}

```

Long Question 2013

Q#3

Write C++ program to evaluate the $\int_1^6 (3x^2 + 1)$ by Simpson's (1/3) or Weddle's Rule due to options 1 or 2 respectively (use n=6). Report the error message for any other option pressed by the user.

Program:

```

#include<iostream>
#include<conio.h>
#include<math.h>
using namespace std;
double long f(double long x)
{
    return (3*x*x+1);
}
main()
{
    int option;
    float a,b,h,n,I;
    float sum=0;
    cout<<"Press 1 for Simpson's rule and Press 2 for wedddle's rule:\t";
    cin>>option;
    cout<<"enter lower limit:\n";
    cin>>a; //1
    cout<<"enter upper limit:\n";
    cin>>b; //6
    cout<<"enter number of intervals:\n";
    cin>>n; //n=6 given
    switch(option)
    {
        case 1:
            cout<<"integration by Simpson's Rule:\n";

```

```

h=(b-a)/n;
sum=f(b)+f(a);
for (int i=1;i<n;i++)
{
    if (i%2==0)
    {
        sum=sum+2*f(a+(i*h));
    }
    else
    {
        sum=sum+4*f(a+(i*h));
    }
}
I=(h/3)*sum;
cout<<"Definite Integration:\t"<<I<<endl;
break;
case 2:
cout<<"integration by Weddle's Rule:\n";
h=(b-a)/n;
sum=f(b)+f(a);
for (int i=1;i<n;i++)
{
    if (i%2==0)
    {
        sum=sum+f(a+(i*h));
    }
    else if(i%3==0)
    {
        sum=sum+6*f(a+(i*h));
    }
    else
    {
        sum=sum+5*f(a+(i*h));
    }
}

```

```

        }

    }

    I=(3*h*sum)/10;

    cout<<"Definite Integration:\t"<<I<<endl;

    break;

    default:

        {

            cout<<" error :Invalid option"<<endl;

            break;

        }

    }

    return 0;

}

```

Q#4

Suppose A & B be 3x3 matrices. Write C++ program which reads in entries of A & B and Prints out the entries of matrix which is

- (i) $A+B$ and (ii) $5A-B^t$

Program:

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    float A[3][3],B[3][3],C[3][3],D[3][3],E[3][3];
    int i,j;
    cout<<"enter the values in matrix A:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cin>>A[i][j];
        }
    }
}

```

```
cout<<"enter the values in matrix B:\n";
```

```
for(i=0;i<3;i++)
```

```
{
```

```
    for(j=0;j<3;j++)
```

```
{
```

```
        cin>>B[i][j];
```

```
}
```

```
}
```

```
cout<<"entered values in matrix A:\n";
```

```
for(i=0;i<3;i++)
```

```
{
```

```
    for(j=0;j<3;j++)
```

```
{
```

```
        cout<<A[i][j]<<"\t";
```

```
}
```

```
cout<<endl;
```

```
}
```

```
cout<<"entered values in matrix B:\n";
```

```
for(i=0;i<3;i++)
```

```
{
```

```
    for(j=0;j<3;j++)
```

```
{
```

```
        cout<<B[i][j]<<"\t";
```

```
}
```

```
cout<<endl;
```

```
}
```

```
for(i=0;i<3;i++)
```

```
{
```

```
    for(j=0;j<3;j++)
```

```
{
```

```
        C[i][j]=A[i][j]+B[i][j];
```

```
        D[i][j]=B[j][i];
```

```
        E[i][j]=5*A[i][j]-D[i][j];
```



```

    }
    }

    cout<<" Sum of A+B is:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cout<<C[i][j]<<"\t";
        }
        cout<<endl;
    }

    cout<<" Sum of 5A-Bt is:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cout<<E[i][j]<<"\t";
        }
        cout<<endl;
    }
    return 0;
}

```

Q#5

Write C++ Program for the simple Harmonic Motion (S.H.M) of a mass attached with a spring using Euler's method under the following conditions:

($g=9.8 \text{ ms}^{-2}$, initial position=zero ,and velocity= 15 ms^{-1} time step 0.1 sec and maximum time 15 sec. $k=1 \text{ N/m}$, $m=1 \text{ Kg}$).

Calculate and Print values for time , Position and velocity. How u can change the same program for forced harmonic motion, Damped Harmonic Motion.(give short comments only).

Program:

```

#include<iostream>

#include<math.h>

#include<iomanip>

using namespace std;

```

```

main()
{
    float x=0,t=0,a,tmax=15,v=15,k=-1,m=1,h=0.1;
    a=(-1)*(k*x)/m; //(F=ma=-kx so we get formula a=-kx/m)
    int p=0;
    float X[1000],T[1000],V[1000],A[1000];
    while (t<=tmax)
    {
        T[p]=t;
        V[p]=v;
        X[p]=x;
        A[p]=a;
        x=x+(x*h); //xf=xi+vt bcz s=x=vt
        v=v+(a*h); //Vf=vi+at
        t=t+h;
        p++; //increment operator
    }
    cout<<setw(16)<<"Time"<<setw(16)<<"Position"<<setw(16)<<"Velocity"<<setw(16)<<"A
cceleration"<<endl;
    for(int i=0;i<p;i++)
    {
        cout<<setw(16)<<T[i]<<setw(16)<<X[i]<<setw(16)<<V[i]<<setw(16)<<A[i]<<endl;
    }
    return 0;
}

```

Short Question 2014

Write C++ program Code Segment for the following:

(a) To calculate equivalent capacitance of two capacitor connected in series

Program:

```

#include<iostream>
using namespace std;
main()
{
    float C,C1,C2,Ceq=0;
    cout<<"Enter the value of capacitor 1:\n";
    cin>>C1;
    cout<<"Enter the value of Capacitor 2:\n";
    cin>>C2;

```



```

        Ceq=(C1*C2)/(C1+C2);
        cout<<"The equivalent capacitance in series combination==:"<<Ceq;
        return 0;
    }

```

(b) To print “x” against “f(X)” for the equation

$$f(x)=25x^3-44x^2-21x+47$$

Program: (simple)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    float x;
    float fx;
    cout<<"enter the value of x:\n";
    cin>>x;
    fx=25*pow(x,3)-44*pow(x,2)-21*x+47;
    cout<<"Value of x="<<x<<" & corresponding function="<<fx;
    return 0;
}

```

Program: (using Array)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int i,j,s;
    cout<<"enter the size of array :'\n";
    cin>>s;
    float x[s],y[s];
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    for (j=0;j<s;j++)
    {
        y[j]= 25*(pow(x[j],3))-44*(pow(x[j],2))-21*x[j]+47;
        cout<<"Values of X is="<<x[j]<<"t and f(x)="<<y[j]<<endl;
    }
}

```

```
return 0;
```

```
}
```

(c) To calculate the factorial of a number (Repeat 2013)

Program:

```
#include<iostream>
using namespace std;
main()
{
    float R,n,Req=0;
    cout<<"Enter the Number of Resistance In Series :\n";
    cin>>n;
    cout<<"Enter the value of Resistance:\n";
    cin>>R;
    for (int i=0;i<n;i++)
    {
        Req=Req+R;
    }
    cout<<"The equivalent resistance in series combination of same resistance =:\n"<<Req;
    return 0;
}
```

(d) To convert from Kilogram to gram using a function ?

Program:

```
#include<iostream>
using namespace std;
float gram(float kg)
{
    float g;
    g=kg/1000;
    return g;
}
main()
{
    float kilo;
    cout<<"enter the value in kilogram:\n";
    cin>>kilo;
    cout<<"print value in gram:="<<gram(kilo);
    return 0;
}
```

Long Question 2014

Q#3 Write C++ program to evaluate the $\int_1^6 \frac{dx}{x^2-1}$ by Trapezoidal or Weddle's Rule due to options 1 or 2 respectively (use n=6). Report the error message for any other option pressed by the user.

Program:

```
#include<iostream>
#include<conio.h>
#include<math.h>
using namespace std;
double long f(double long x)
{
    return (1/(x*x-1));
}
```

```

}
main()
{
    int option;
    float a,b,h,n,I;
    float sum=0;
    cout<<"Press 1 for Trapezoidal and Press 2 for wedddle's rule:\t";
    cin>>option;
    cout<<"enter lower limit:\n";
    cin>>a; //1
    cout<<"enter upper limit:\n";
    cin>>b; //6
    cout<<"enter number of intervals:\n";
    cin>>n; //n=6 given
    switch(option)
    {
        case 1:
            cout<<"integration by Trapezoidal Rule:\n";
            h=(b-a)/n;
            sum=f(b)+f(a);
            for (int i=1;i<n;i++)
            {
                sum=sum+2*f(a+(i*h));
            }
            I=(h/2)*sum;
            cout<<"Definite Integration:\t"<<I<<endl;
            break;
        case 2:
            cout<<"integration by Weddle's Rule:\n";
            h=(b-a)/n;
            sum=f(b)+f(a);
            for (int i=1;i<n;i++)
            {

```

```

        if (i%2==0)
        {
            sum=sum+f(a+(i*h));
        }
        else if(i%3==0)
        {
            sum=sum+6*f(a+(i*h));
        }
        else
        {
            sum=sum+5*f(a+(i*h));
        }
    }
    I=(3*h*sum)/10;
    cout<<"Definite Integration:\t"<<I<<endl;
    break;
default:
{
    cout<<" error :Invalid option"<<endl;
    break;
}
}
return 0;
}

```

Q#4

Suppose A & B be 3x3 matrices. Write C++ program which reads in entries of A & B and Prints out the entries of matrix which is

- (i) $A-B$ and (ii) $C=B^t$

Program:

```

#include<iostream>

#include<math.h>

using namespace std;

main()

```

```

{

float A[3][3],B[3][3],C[3][3],D[3][3];

int i,j;

cout<<"enter the values in matrix A:\n";

for(i=0;i<3;i++)

{

    for(j=0;j<3;j++)

    {

        cin>>A[i][j];

    }

}

cout<<"enter the values in matrix B:\n";

for(i=0;i<3;i++)

{

    for(j=0;j<3;j++)

    {

        cin>>B[i][j];

    }

}

cout<<"entered values in matrix A:\n";

for(i=0;i<3;i++)

{

    for(j=0;j<3;j++)

    {

        cout<<A[i][j]<<"\t";

    }

}

cout<<endl;

}

cout<<"entered values in matrix B:\n";

for(i=0;i<3;i++)

{

    for(j=0;j<3;j++)

    {

```

```

        cout<<B[i][j]<<"\t";

    }
    cout<<endl;
}
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        D[i][j]=A[i][j]-B[i][j];
        C[i][j]=B[j][i];
    }
}
cout<<" Result of A-B is:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<D[i][j]<<"\t";
    }
    cout<<endl;
}
cout<<" Sum of C=Bt is:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<C[i][j]<<"\t";
    }
    cout<<endl;
}
return 0;

```

}

Q#5 (a)

Find the roots of the equation " $3x - \cos x - 5$ " using simple iterative method using $X_0 = 1.5$. Write C++ Program to implement the method correct to 2 decimal places?

Program:

```
#include<iostream>
#include<conio.h>
#include<math.h>
#include<iomanip>
using namespace std;
float f(float x)
{
    return (0.333*(cos(x)+5));
}
main()
{
    float E,X0,X1,X2,x;
    int n,i,flag=1;
    cout<<" enter an educated guess ";
    cin>>n;
    cout<<" enter value for X0 ";
    cin>>X0;
    cout<<" enter accuracy ";
    cin>>E;
    cout<<" iteration \t flag \t root ";
    cout<<setw(8)<<"iteration"<<setw(16)<<"x(n+1)"<<setw(16)<<"x(n)"<<setw(16)<<"x(n+1)-x(n)"<<endl;
    for( i=1; i<n&&flag; i++ )
    {
        X1=f(X0);
        x=X1-X0;
        if(x<0)
        {
            x=-x;
        }
    }
```

```

if(x<E)
{
flag=0;
}
else
{
cout<<setw(8)<<i<<setw(16)<<X1<<setw(16)<<X0<<setw(16)<<x<<endl;
X0=X1;
}
}
if(flag==1)
{
cout<<" solution does not exist ";
}
else
{
    cout<<" after "<<i-1<<" iteration root= "<<X1<<endl;
}
return 0;
}

```

Q#5(b) Write C++ program which reads in a number and display its table. Implement your program for 10 iterations?

Program:

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int n,i;
    cout<<"enter the number:\t";
    cin>>n;
    for(i=1;i<=10;i++)

```

```

{
cout<<n<<"X"<<i<<"="<<n*i<<endl;
}

return 0;
}

```

Short Question 2015

Write C++ program Code Segment for the following:

- (a) To calculate equivalent capacitance of two capacitor connected in parallel

Program:

```

#include<iostream>
using namespace std;
main()
{
    float C,C1,C2,Ceq=0;
    cout<<"Enter the value of capacitor 1:\n";
    cin>>C1;
    cout<<"Enter the value of Capacitor 2:\n";
    cin>>C2;
    Ceq=C1+C2;
    cout<<"The equivalent capacitance in parallel combination==:"<<Ceq;
    return 0;
}

```

- (b) To print “x & y” against “f(X,Y)” for the equation

$$f(x) = 12x^2 - 4y^2$$

Program: (simple)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    float x,y;
    float fx;
    cout<<"enter the value of x:\n";
    cin>>x;
    cout<<"enter the value of y:\n";
    cin>>y;
    fx=12*pow(x,2)-4*pow(y,2);
    cout<<"Value of (x,y):("<<x<<","<<y<<") & corresponding function="<<fx;
    return 0;
}

```

Program: (using array)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int i,j,k,s;
    cout<<"enter the size of array :\n";
    cin>>s;
    float x[s],y[s],z[s];
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    for (j=0;j<s;j++)
    {
        cout<<"Enter The values of y"<<j<<endl;
        cin>>y[j];
    }
    for (k=0;k<s;k++)
    {
        z[k]=12*(pow(x[k],2))-4*(pow(y[k],2));
        cout<<" values of (x,y):("<<x[k]<<","<<y[k]<<")"<<" and f(x,y)="<<z[k]<<endl;
    }
    return 0;
}

```

(c) To calculate the table of number 5

Program:

```

#include<iostream>
using namespace std;
main()
{
    int i,n,a;

    n=5;
    cout<<"Table of Number 5 is:"<<endl;
    for (int i=1;i<=10;i++)
    {
        a=n*i;
        cout<<n<<"x"<<i<<"="<<a<<endl;
    }
}

```

```
return 0;
```

```
}
```

(d) To convert temperature from Celsius to fahrenheit scale using a function ?

Program:

```
#include<iostream>
```

```
using namespace std;
```

```
float fah (float c)
```

```
{
```

```
float f;
```

```
f=(9*c)/5+32;
```

```
return f;
```

```
}
```

```
main()
```

```
{
```

```
float c;
```

```
cout<<"enter the celcius temperature:\n";
```

```
cin>>c;
```

```
cout<<"Temperaturein fahrenheit:="<<fah(c);
```

```
return 0;
```

```
}
```

Long Question 2015

Q#3 Write C++ program to evaluate the $\int_1^6 (2x^2 - 1) dx$ by Trapezoidal or Weddle's Rule due to options 1 or 2 respectively (use n=6). Report the error message for any other option pressed by the user.

Program:

```
#include<iostream>
```

```
#include<conio.h>
```

```
#include<math.h>
```

```
using namespace std;
```

```
double long f(double long x)
```

```
{
```

```
return (2*x*x-1);
```

```
}
```

```
main()
```

```
{
```

```

int option;
float a,b,h,n,I;
float sum=0;
cout<<"Press 1 for Trapezoidal and Press 2 for
weddle's rule:\t";

cin>>option;
cout<<"enter lower limit:\n";
cin>>a; //1
cout<<"enter upper limit:\n";
cin>>b; //6
cout<<"enter number of intervals:\n";
cin>>n; //n=6 given
switch(option)
{
    case 1:
        cout<<"integration by Trapezoidal Rule:\n";
        h=(b-a)/n;
        sum=f(b)+f(a);
        for (int i=1;i<n;i++)
        {
            sum=sum+2*f(a+(i*h));
        }
        I=(h/2)*sum;
        cout<<"Definite Integration:\t"<<I<<endl;
        break;
    case 2:
        cout<<"integration by Weddle's Rule:\n";
        h=(b-a)/n;

```



```

sum=f(b)+f(a);
for (int i=1;i<n;i++)
{
    if (i%2==0)
    {
        sum=sum+f(a+(i*h));
    }
    else if(i%3==0)
    {
        sum=sum+6*f(a+(i*h));
    }
    else
    {
        sum=sum+5*f(a+(i*h));
    }
}
I=(3*h*sum)/10;
cout<<"Definite Integration:\t"<<I<<endl;
break;
default:
{
    cout<<" error :Invalid option"<<endl;
    break;
}
}
return 0;

```

}

Q#4

Suppose A & B be 3x3 matrices. Write C++ program which reads in entries of A & B and Prints out the entries of matrix which is

(i) $K=7A-5B$ and (ii) $C=Ax B^t$

Program:

```
#include<iostream>

#include<math.h>

using namespace std;

main()
{
    float A[3][3],B[3][3],C[3][3],K[3][3],E[3][3];
    int i,j,l,sum;
    cout<<"enter the values in matrix A:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cin>>A[i][j];
        }
    }
    cout<<"enter the values in matrix B:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cin>>B[i][j];
        }
    }
    cout<<"entered values in matrix A:\n";
    for(i=0;i<3;i++)
```

```

{
    for(j=0;j<3;j++)
    {
        cout<<A[i][j]<<"\t";
    }
    cout<<endl;
}
cout<<"entered values in matrix B:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<B[i][j]<<"\t";
    }
    cout<<endl;
}
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        K[i][j]=7*A[i][j]-5*B[i][j];
        E[i][j]=B[j][i];
    }
}
cout<<" Sum of k=7A-5B is:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)

```

```

{
cout<<K[i][j]<<"\t";
}
cout<<endl;
}
for (i=0;i<3;i++)
{
    for (j=0;j<3;j++)
    {
        sum=0;
        for(l=0;l<3;l++)
            sum=sum+A[i][l]*E[l][j];
        C[i][j]=sum;
    }
}
cout<<"Result of C=A*B'\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<C[i][j]<<"\t";
    }
    cout<<endl;
}

return 0;
}

```

Q#5 (a)

Write C++ Program for the simple Harmonic Motion (S.H.M) of a mass attached with a spring using Euler's method under the following conditions:

($g=9.8 \text{ ms}^{-2}$, initial position =zero ,and velocity= 15 ms^{-1} time step 0.1 sec and maximum time 15 sec. $k=1 \text{ N/m}$, $m=1 \text{ Kg}$).

Calculate and Print values for time , Position and velocity. How u can change the same program for forced harmonic motion, Damped Harmonic Motion.(give short comments only).

Program:

REPEAT 2013 Question #5 Long

Q#5(b)

Write C++ Program which reads in a number and display its factorial.

Implement your Program for 10 iterations?

Program: (same Number 10 Times)

```
#include<iostream>
#include<math.h>
using namespace std;
unsigned long fact(int n)
{
    int fact=1;
    for(int i=1;i<=n;i++)
        fact=fact*i;
    return fact;
}
main()
{
    int c,n;
    cout<<"enter positive number :n";
    cin>>n;
    for(c=1;c<=10;c++)
        cout<<"factorial of"<<c<<" iteration: "<<fact(n)<<endl;
    return 0;
}
```

Program: (Different 10 Number) { 1st is correct }

```
#include<iostream>
#include<math.h>
using namespace std;
unsigned long fact(int n)
{
    int fact=1;
    for(int i=1;i<=n;i++)
        fact=fact*i;
    return fact;
}
main()
{
    int n;
    for(n=1;n<=10;n++)
        cout<<"factorial of"<<n<<"="<<fact(n)<<endl;
    return 0;
}
```

Short Question 2016

Write C++ program Code Segment for the following:

- (a) To calculate equivalent capacitance of two capacitor connected in parallel (Repeat 2015)**

Program:

```
#include<iostream>
using namespace std;
main()
{
    float C,C1,C2,Ceq=0;
    cout<<"Enter the value of capacitor 1:\n";
    cin>>C1;
    cout<<"Enter the value of Capacitor 2:\n";
    cin>>C2;
    Ceq=C1+C2;
    cout<<"The equivalent capacitance in parallel combination==:"<<Ceq;
    return 0;
}
```


(b) To print “x” and “f(X)” for the equation

$$f(x) = \frac{(x^2 - 2x + 4)}{15}$$

Program: (simple)

```
#include<iostream>
#include<math.h>
using namespace std;
main()
{
    float x;
    long double fx;
    cout<<"enter the value of x:\n";
    cin>>x;
    fx=(pow(x,2)-2*x+4)/15;
    cout<<"Value of x:"<<x<<" & corresponding function="<<fx;
    return 0;
}
```

Program: (using array)

```
#include<iostream>
#include<math.h>
#include<stdio.h>
using namespace std;
main()
{
    int i,j,s;
    cout<<"enter the size of array :\n";
    cin>>s;
    float x[s],y[s];
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    for (j=0;j<s;j++)
    {
        y[j]=(pow(x[j],2)-2*x[j]+4)/15;
        cout<<" values of x="<<x[j]<<" and f(x)="<<y[j]<<endl;
    }
}
```

```
return 0;
```

```
}
```

(c) To calculate the factorial of a number read from the user ?

Program: (Repeat 2013)

```
#include<iostream>
using namespace std;
main()
{
    unsigned long fact=1;
    int i,n;
    cout<<"Enter the number:\n";
    cin>>n; //any number whose factorial you want to find
    for(i=1;i<=n;i++)
    {
        fact=fact*i;
    }
    cout<<"factorial of number is="<<fact;
    return 0;
}
```

(d) To convert temperature from Celsius to fahrenheit scale using a function ?

Program: (Repeat 2015)

```
#include<iostream>
using namespace std;
float fah (float c)
{
    float f;
    f=(9*c)/5+32;
    return f;
}
main()
{
    float c;
    cout<<"enter the celcius temperature:\n";
    cin>>c;
    cout<<"Temperature in fahrenheit:="<<fah(c);
    return 0;
}
```

Long Question 2016

Q#3 Write C++ program to evaluate the $\int_1^6 \sqrt{x} dx$ by Trapezoidal or Simpson's Rule due to options 1 or 2 respectively (use n=6). Report the error message for any other option pressed by the user.

Program:

```
#include<iostream>
#include<conio.h>
```

```

#include<math.h>

using namespace std;

double long f(double long x)
{
    return (sqrt(x));
}

main()
{
    int option;
    float a,b,h,n,I;
    float sum=0;
    cout<<"Press 1 for Trapezoidal and Press 2 for Simpson's Rule:\t";
    cin>>option;
    cout<<"enter lower limit:\n";
    cin>>a; //1
    cout<<"enter upper limit:\n";
    cin>>b; //6
    cout<<"enter number of intervals:\n";
    cin>>n; //n=6 given
    h=(b-a)/n;
    sum=f(b)+f(a);
    switch(option)
    {
        case 1:
            cout<<"integration by Trapezoidal Rule:\n";
            for (int i=1;i<n;i++)
            {
                sum=sum+2*f(a+(i*h));
            }
            I=(h/2)*sum;
            cout<<"Definite Integration:\t"<<I<<endl;
            break;
        case 2:

```

```

        cout<<"integration by Simpson's Rule:\n";
        for (int i=1;i<n;i++)
        {
            if (i%2==0)
            {
                sum=sum+2*f(a+(i*h));
            }
            else
            {
                sum=sum+4*f(a+(i*h));
            }
        }
        I=(h/3)*sum;
        cout<<"Definite Integration:\t"<<I<<endl;
        break;
        default:
        {
            cout<<" error :Invalid option"<<endl;
            break;
        }
    }
    return 0;
}

```

Q#4 Suppose A & B be 3x3 matrices. Write C++ program which reads in entries of A & B and Prints out the entries of matrix which is

- (i) $A \times B$ and (ii) $4A + 2B^t$

Program:

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int A[3][3],B[3][3],C[3][3],D[3][3],E[3][3];

```

```

int i,j,k,sum;
cout<<"enter the values in matrix A:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cin>>A[i][j];
    }
}
cout<<"enter the values in matrix B:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cin>>B[i][j];
    }
}
cout<<"entered values in matrix A:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<A[i][j]<<"\t";
    }
    cout<<endl;
}
cout<<"entered values in matrix B:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<B[i][j]<<"\t";
    }
}

```

```

cout<<endl;
}
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        sum=0;
        for(k=0;k<3;k++)
        {
            sum=sum+A[i][k]*B[k][j];
            C[i][j]=sum;
        }
    }
}
cout<<"result AxB is:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<C[i][j]<<"\t";
    }
    cout<<endl;
}

for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        D[i][j]=B[j][i];
        E[i][j]=4*A[i][j]+2*D[i][j];
    }
}

cout<<"Result of 4A+2B: \n";
for(i=0;i<3;i++)

```



```

{
    for(j=0;j<3;j++)
    {
        cout<<E[i][j]<<"\t";
    }
    cout<<endl;
}
return 0;
}

```

Q#5(a)

Write C++ Program to **determine** equivalent resistance of n resistances Connected in Series (READ RESISTANCE VALUE FROM USER).

Program:

```

#include<iostream>
using namespace std;
main()
{
    float R,Req;
    int n;
    cout<<"Enter the value of resistance of R:=\t";
    cin>>R;
    cout<<"Enter Total resistacnce connected in series:=\t";
    cin>>n;
    Req=n*R;
    cout<<"Equivalent resistance of"<<" "<<n<<" "<<"Resistance is=\t";
    cout<<Req;
    return 0;
}

```

Q#5(b)

What is an Array? Write down the general syntax for the usage of an array, if “a” ,”b” and “c” are the three linear arrays,

Write C++ Program to calculate the $c = a^2 - b^2$, Initialize “a” by ten odd numbers and “b” by ten even numbers. Implement your program using arrays?

Program:

Array:

An array contains Multiple objects of identical types stored sequentially in memory. The Individual objects in an array, referred to as “**Array Elements**”, can be addressed using a number ,called “**Index or Subscript .**”

Syntax:

The general syntax of One Dimensional Linear Array is below:

Type Name [Count]

Program:

```
#include<iostream>
#include<iomanip>
using namespace std;
main()
{
    int a[10]={1,3,5,7,9,11,13,15,17,19};
    int b[10]={2,4,6,8,10,12,14,16,18,20};
    int c[9],i;
    for(i=0;i<10;i++)
    {
        c[i]=a[i]*a[i]-b[i]*b[i];
    }
    for(i=0;i<10;i++)
    cout<<setw(10)<<a[i]<<setw(10)<<b[i]<<setw(20)<<c[i]<<endl;
    return 0;
}
```

Short Question 2017:

Write C++ program Code Segment for the following:

(a) To print table of number enter by user

Program:

```
#include<iostream>
using namespace std;
main()
{
```

```

int n,i;
cout<<"Enter the number:\n";
cin>>n;
cout<<"table of number "<<n<<" is:\n";
for (i=1;i<=10;i++)
{
cout<<n<<"x"<<i<<"="<<n*i<<endl;
}
return 0;
}

```

(b) To print “x” against “f(X)” for the equation

$$f(x)=7\sqrt{(3x^2 - 2x - 22)}$$

and calculate the maximum and minimum of f(x)

Program:

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int i,j,k,l,s;
    cout<<"enter the size of array :\n";
    cin>>s;
    float x[s],y[s];
    float max,min;
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    for (j=0;j<s;j++)
    {
        y[j]= 7*sqrt(3*(pow(x[j],2))-2*x[j]-22);
        cout<<"Values of X is="<<x[j]<<"\t and f(x)="<<y[j]<<endl;
    }
    max=y[0];
    for(k=0;k<s;k++)
    {
        if (max<y[k])
            max=y[k];
    }
    cout<<"Maximum value of f(x) is="<<max<<endl;
}

```

```

min=y[0];
for(l=0;l<s;l++)
{
if (min>y[l])
min=y[l];
}

cout<<"Minimum value of f(x) is"<<min<<endl;

return 0;
}

```

(Above Programme New Method)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int i,j,k,max,min,s;
    cout<<"enter the size of array :'\n";
    cin>>s;
    float x[s],y[s],fx[s];
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    for (j=0;j<s;j++)
    {
        y[j]=(3*x[j]*x[j]-2*x[j]-22);
        fx[j]=7*sqrt(y[j]);
        cout<<"Values of X is"<<x[j]<<"\t and f(x)=""<<fx[j]<<endl;
    }
    max=fx[s];

for( j=0; j<s; j++ )
{
if(max<fx[j])
max=fx[j];
}

cout<<" the max of F(x) is ""<<max<<endl;
min=fx[0];
for( j=0; j<s; j++ )

```

```

{
if(min>fx[j])
min=fx[j];
}

cout<<" the min of F(x) is "<<min<<endl;

return 0;
}

```

(c) To read the (x,y) values from user and calculate polar coordinates (r,θ)?

Program: (Simple)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int x,y;
    float t,r,theeta;
    cout<<"enter the value of X & y:\n";
    cin>>x>>y;
    r=sqrt(x*x+y*y);
    t=y/x;
    theeta=atan(t);
    cout<<"value of (x,y)=("<<x<<","<<y<<") and (r,theeta)=("<<r<<","<<theeta<<")"<<endl;
    return 0;
}

```

Program: (using function)

```

#include<iostream>
#include<math.h>
using namespace std;
float r(float x,float y)
{
    return (sqrt(pow(x,2)+pow(y,2)));
}
float theta(float x,float y)
{
    return (atan(y/x));
}
main()
{
    float x,y;
    cout<<"enter the x coordinates:\n";
    cin>>x;
    cout<<"enter the y coordinates:\n";
    cin>>y;
    cout<<"values of (x,y)="<<("("<<x<<","<<y<<")"<<"and (r,θ)=("<<r(x,y)<<","<<theta(x,y)<<")"<<endl;
    return 0;
}

```

Long Question 2017::

Q#3 Write C++ program to evaluate the $\int_1^6 \frac{x^2+4}{5} dx$ by Trapezoidal or Weddle's Rule due to options 1 or 2 respectively (use n=6). Report the error message for any other option pressed by the user.

Program:

```

#include<iostream>

#include<conio.h>

#include<math.h>

```

```

using namespace std;

double long f(double long x)
{
    return ((x*x+4)/5);
}

main()
{
    int option;
    float a,b,h,n,I;
    float sum=0;
    cout<<"Press 1 for Trapezoidal and Press 2 for wedddle's rule:\t";
    cin>>option;
    cout<<"enter lower limit:\n";
    cin>>a; //1
    cout<<"enter upper limit:\n";
    cin>>b; //6
    cout<<"enter number of intervals:\n";
    cin>>n; //n=6 given
    switch(option)
    {
        case 1:
            cout<<"integration by Trapezoidal Rule:\n";
            h=(b-a)/n;
            sum=f(b)+f(a);
            for (int i=1;i<n;i++)
            {
                sum=sum+2*f(a+(i*h));
            }
            I=(h/2)*sum;
            cout<<"Definite Integration:\t"<<I<<endl;
            break;
        case 2:
            cout<<"integration by Weddle's Rule:\n";

```



```

h=(b-a)/n;
sum=f(b)+f(a);
for (int i=1;i<n;i++)
{
    if (i%2==0)
    {
        sum=sum+f(a+(i*h));
    }
    else if(i%3==0)
    {
        sum=sum+6*f(a+(i*h));
    }
    else
    {
        sum=sum+5*f(a+(i*h));
    }
}
I=(3*h*sum)/10;
cout<<"Definite Integration:\t"<<I<<endl;
break;
default:
{
    cout<<" error :Invalid option"<<endl;
    break;
}
}
return 0;
}

```

Q#4 Suppose A & B be 3x3 matrices. Write C++ program which reads in entries of A & B and Prints out the entries of matrix which is

- (i) $C=A \times B$ and (ii) $2A^t + 3B$

Program:

```
#include<iostream>
```

```

#include<math.h>

using namespace std;

main()
{
    int A[3][3],B[3][3],C[3][3],D[3][3],E[3][3];
    int i,j,k,sum;
    cout<<"enter the values in matrix A:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cin>>A[i][j];
        }
    }
    cout<<"enter the values in matrix B:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cin>>B[i][j];
        }
    }
    cout<<"entered values in matrix A:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cout<<A[i][j]<<"\t";
        }
        cout<<endl;
    }
    cout<<"entered values in matrix B:\n";

```

```

for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<B[i][j]<<"\t";
    }
    cout<<endl;
}

for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        sum=0;
        for(k=0;k<3;k++)
        {
            sum=sum+A[i][k]*B[k][j];
            C[i][j]=sum;
        }
    }
}

cout<<"result C=AxB is:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<C[i][j]<<"\t";
    }
    cout<<endl;
}

for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)

```

```

        {
            D[i][j]=A[j][i];
            E[i][j]=2*D[i][j]+3*B[i][j];
        }
    }
    cout<<"Result of 2A'+3B: \n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cout<<E[i][j]<<"\t";
        }
        cout<<endl;
    }
    return 0;
}

```

Q#5 (a)

Find the roots of the equation “ $3x - \cos x - 5$ ” using simple iterative method using $X_0 = 1.5$. Write C++ Program to implement the method correct to 2 decimal places?

Program:

Repeat 2014 Q#5 (a)

Q#5 (b)

Write C++ Program for Conversion of

- (i) Temperature From “F” to “C”?
- (ii) Length from “m” to “km”?

Program: (Temperature From “F” to “C”)

SIMPLE:

```

#include<iostream>
using namespace std;
main()
{

```

```

    float f,c;

```

```

    cout<<"enter the temperatue in fahrenheit:\t";

    cin>>f;

    c=5*(f-32)/9;

    cout<<"Teperature in celcuis:\t"<<c<<endl;

    return 0;

}

```

USING FUNCTION:

```

#include<iostream>

using namespace std;

float c(float f)
{
    return (5*(f-32)/9);
}

main()
{
    float f;

    cout<<"enter the temperatue in fahrenheit:\t";

    cin>>f;

    cout<<"Teperature in celcuis:\t"<<c(f)<<endl;

    return 0;

}

```

Program: (Length from “m” to “km”)

SIMPLE:

```

#include<iostream>

using namespace std;

main()
{
    float m,km;

    cout<<"enter the distance in metre:\t";

    cin>>m;

    km=(m/1000);

    cout<<"distance in kilometer:\t"<<km<<endl;
}

```

```

    return 0;
}

```

USING FUNCTION:

```

#include<iostream>

using namespace std;

int km(int meter)
{
    return (meter/1000);
}

main()
{
    float m;

    cout<<"enter the distance in metre:\t";
    cin>>m;

    cout<<"distance in kilometer:\t"<<km(m)<<endl;

    return 0;
}

```

Short Question 2018

Write C++ program Code Segment for the following:

- (a) To calculate and print equivalent capacitance of two capacitor connected in series read capacitance from user? (Repeat-2014 a)

Program:

```

#include<iostream>
using namespace std;
main()
{
    float C,C1,C2,Ceq=0;
    cout<<"Enter the value of capacitor 1:\n";
    cin>>C1;
    cout<<"Enter the value of Capacitor 2:\n";
    cin>>C2;
    Ceq=(C1*C2)/(C1+C2);
    cout<<"The equivalent capacitance in series combination==:"<<Ceq;
    return 0;
}

```

- (b) To print “x & y” against “f(X,Y)” for the equation

$$f(x) = x^2 + 77xy - y^2$$

Program: (simple)

```

#include<iostream>

#include<math.h>

using namespace std;

```



```

main()
{
    float x,y;
    float fx;
    cout<<"enter the value of x:\n";
    cin>>x;
    cout<<"enter the value of y:\n";
    cin>>y;
    fx=pow(x,2)+77*x*y-pow(y,2);
    cout<<"Value of (x,y):("<<x<<","<<y<<") & corresponding function="<<fx;
    return 0;
}

```

Program: (using array)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int i,j,k,s;
    cout<<"enter the size of array :\n";
    cin>>s;
    float x[s],y[s],z[s];
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    for (j=0;j<s;j++)
    {
        cout<<"Enter The values of y"<<j<<endl;
        cin>>y[j];
    }
    for (k=0;k<s;k++)
    {
        z[k]=(pow(x[k],2))+77*x[k]*y[k]-(pow(y[k],2));
    }
}

```

```

        cout<<" values of (x,y):("<<x[k]<<","<<y[k]<<")"<<" and f(x,y)="<<z[k]<<endl;
    }

    return 0;
}

```

(c) For Minimum Value of F(x) see Below Program

Program:

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    int i,j,k,s;
    cout<<"enter the size of array :'\n";
    cin>>s;
    float x[s],y[s],z[s],min ;
    for (i=0;i<s;i++)
    {
        cout<<"Enter The values of X"<<i<<endl;
        cin>>x[i];
    }
    for (j=0;j<s;j++)
    {
        cout<<"Enter The values of y"<<j<<endl;
        cin>>y[j];
    }
    for (k=0;k<s;k++)
    {
        z[k]=(pow(x[k],2))+77*x[k]*y[k]-(pow(y[k],2));
        cout<<" values of (x,y):("<<x[k]<<","<<y[k]<<")"<<" and f(x,y)="<<z[k]<<endl;
    }
    min=z[0];
    for(k=0;k<s;k++)
    {
        if (min>z[k])
            min=z[k];
    }
    cout<<"minimum Value of f(x,y)= "<<min<<endl;
    return 0;
}

```

Long Question 2018

Q#3 Write C++ program to evaluate the $\int_1^6 (x^2 + 4\sqrt{x} - 7) dx$ by Trapezoidal or Simpson's (1/3)Rule due to options 1or 2 respectively (use n=6). Report the error message for any other option pressed by the user.

Program:

```

#include<iostream>

#include<iostream>

#include<conio.h>

#include<math.h>

using namespace std;

double long f(double long x)

```

```

{
    return (pow(x,2)+4*sqrt(x)-7);
}

main()
{
    int option;
    float a,b,h,n,I;
    float sum=0;
    cout<<"Press 1 for Trapezoidal and Press 2 for Simpson's rule:\t";
    cin>>option;
    cout<<"enter lower limit:\n";
    cin>>a; //1
    cout<<"enter upper limit:\n";
    cin>>b; //6
    cout<<"enter number of intervals:\n";
    cin>>n; //n=6 given
    h=(b-a)/n;
    sum=f(b)+f(a);
    switch(option)
    {
        case 1:
            cout<<"integration by Trapezoidal Rule:\n";
            for (int i=1;i<n;i++)
            {
                sum=sum+2*f(a+(i*h));
            }
            I=(h/2)*sum;
            cout<<"Definite Integration:\t"<<I<<endl;
            break;
        case 2:
            cout<<"integration by Simpson's Rule:\n";

```

```

for (int i=1;i<n;i++)
{
    if (i%2==0)
    {
        sum=sum+2*f(a+(i*h));
    }
    else
    {
        sum=sum+4*f(a+(i*h));
    }
}
I=(h/3)*sum;
cout<<"Definite Integration:\t"<<I<<endl;
break;
default:
{
    cout<<" error :Invalid option"<<endl;
    break;
}
}
return 0;
}

```

Q#4 Suppose A & B be 3x3 matrices. Write C++ program which reads in entries of A & B and Prints out the entries of matrix which is

- (i) $C = \frac{(11Ax4B)}{44}$ and (ii) $D = A - B^t + 3$
- (ii) Square of the elements of matrix B
- (iii) To print out diagonal elements of C
- (iv) Max of the element of matrix A

Program:

```
#include<iostream>
```

```
#include<math.h>
```

```

using namespace std;

main()
{
    int A[3][3],B[3][3],C[3][3],D[3][3],E[3][3],F[3][3];
    int r,c,k,sum,max;
    cout<<"enter the values in matrix A:\n";
    for(r=0;r<3;r++)
    {
        for(c=0;c<3;c++)
        {
            cin>>A[r][c];
        }
    }
    cout<<"enter the values in matrix B:\n";
    for(r=0;r<3;r++)
    {
        for(c=0;c<3;c++)
        {
            cin>>B[r][c];
        }
    }
    cout<<"entered values in matrix A:\n";
    for(r=0;r<3;r++)
    {
        for(c=0;c<3;c++)
        {
            cout<<A[r][c]<<"\t";
        }
    }
    cout<<endl;
}
cout<<"entered values in matrix B:\n";
for(r=0;r<3;r++)
{

```

```

        for(c=0;c<3;c++)
        {
            cout<<B[r][c]<<"\t";
        }
        cout<<endl;
    }

    for(r=0;r<3;r++)
    {
        for(c=0;c<3;c++)
        {
            sum=0;
            for(k=0;k<3;k++)
            {
                sum=sum+((11*A[r][k])*(4*B[k][c]))/44;
                C[r][c]=sum;
            }
        }
        cout<<"result 11Ax4B/44 is:\n";
        for(r=0;r<3;r++)
        {
            for(c=0;c<3;c++)
            {
                cout<<C[r][c]<<"\t";
            }
            cout<<endl;
        }

        for(r=0;r<3;r++)
        {
            for(c=0;c<3;c++)
            {
                D[r][c]=A[r][c]-B[c][r]+3;
                F[r][c]=pow(B[r][c],2);
            }
        }
    }

```



```

    }
}

cout<<"Result of A-B'+3: \n";
for(r=0;r<3;r++)
{
    for(c=0;c<3;c++)
    {
        cout<<D[r][c]<<"\t";
    }
    cout<<endl;
}

cout<<"square of matrix B;\n";
for(r=0;r<3;r++)
{
    for(c=0;c<3;c++)
    {
        cout<<F[r][c]<<"\t";
    }
    cout<<endl;
}

cout<<"Diagonal Elements matrix C;\n";
for(r=0;r<3;r++)
{
    for(c=0;c<3;c++)
    {
        if(r==c)
            cout<<C[r][c]<<"\t";
    }
    cout<<endl;
}

max=0;
for(r=0;r<3;r++)
{

```

```

        for(c=0;c<3;c++)
        {
            if(max<A[r][c])
            max=A[r][c];
        }
    }

    cout<<"maximum element of Matrix Ais:="<<t"<<max<<endl;

    return 0;

}

```

Q#5 (a)

Write C++ Program for the simple Harmonic Motion (S.H.M) of a mass attached with a spring using Euler's method under the following conditions:

($g=9.8 \text{ ms}^{-2}$, initial position=zero ,and velocity= 15 ms^{-1} time step 0.1 sec and maximum time 15 sec. $k=1 \text{ N/m}$, $m=1 \text{ Kg}$).

Calculate and Print values for time , Position and velocity. How u can change the same program for forced harmonic motion, Damped Harmonic Motion.(give short comments only).

Program:

REPEAT 2013 Question #5 Long

Q#5 (b)

Write a function circle **0** which reads in radius value and print out area and circumference values of circle .Execute the program iteratively for 5 Values.

Program:

```

#include<iostream>

#include<math.h>

using namespace std;

float area(int r)
{
    return (3.14*r*r);
}

float circm(int r)
{
    return (2*3.14*r);
}

```

```

main()
{
    int r,i,j;
    cout<<"enter the radius of circle:\n";
    cin>>r;
    for(i=0;i<5;i++)
    {
        cout<<"area of circle is:=\t"<<area(r)<<endl;
    }
    for(j=0;j<5;j++)
    {
        cout<<"circumference of circle is:=\t"<<circm(r)<<endl;
    }
    return 0;
}

```

Short Question 2019

Short:

Write C++ program Code Segment for the following:

- (a) To calculate equivalent of two resistance connected in series?**

Program:

```

#include<iostream>
using namespace std;
main()
{
    float R1,R2,Req;
    cout<<"Enter the value of Resistance 1:\n";
    cin>>R1;
    cout<<"Enter the value of Resistance 2:\n";
    cin>>R2;
    Req=R1+R2;
    cout<<"The equivalent resistance in series combination of same resistance =:\n"<<Req;
    return 0;
}

```

- (b) To print “x” against “f(X)” for the equation**

$$f(x) = \frac{5}{\sqrt{(2x^3 - 4x^2 - 2x + 4)}}$$

Program: (simple)

```

#include<iostream>
#include<math.h>
using namespace std;
main()

```

```

{

    float x,y;

    float fx;

    cout<<"enter the value of x:\n";

    cin>>x;

    y=2*pow(x,3)-4*pow(x,2)-2*x+4;

    fx=5/sqrt(y);

    cout<<"Value of x:"<<x<<" & corresponding function="<<fx;

    return 0;

}

```

Program: (using array)

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{

    int i,j,k,s;

    cout<<"enter the size of array :n";

    cin>>s;

    float x[s],y[s],z[s];

    for (i=0;i<s;i++)
    {

        cout<<"Enter The values of X"<<i<<endl;

        cin>>x[i];

    }

    for (j=0;j<s;j++)
    {

        y[j]= 2*(pow(x[j],3))-4*(pow(x[j],2))-2*x[j]+4;

        z[j]=5/sqrt(y[j]);

        cout<<"Values of X is"<<x[j]<<"t and f(x)="<<z[j]<<endl;

    }

    return 0;

}

```

(C) To Calculate and display the table of number ?

Program: (Repeat 2017 a)

```

#include<iostream>
using namespace std;
main()
{

    int n,i;

    cout<<"Enter the number:\n";

    cin>>n;

```

```

cout<<"table of number ""<<n<<" 'is:\n";
for (i=1;i<=10;i++)
{
cout<<n<<"x"<<i<<"="<<n*i<<endl;
}
return 0;
}

```

(d) To convert from gram to kilogram using a function ?

Program:

```

#include<iostream>
using namespace std;
float kg(float gram)
{
return (gram/1000);
}
main()
{
float g;
cout<<"enter the value in gram:\n";
cin>>g;
cout<<"print value in kilogram:="<<kg(g);
return 0;
}

```

Long Question 2019

Q#3 Write C++ program to evaluate the $\int_1^6 \frac{1}{x^3-5} dx$ by Simpson's (1/3) Rule or by Trapezoidal Rule due to options 1or 2 respectively (use n=6). Report the error message for any other option pressed by the user.

Program:

```

#include<iostream>
#include<conio.h>
#include<math.h>
using namespace std;
double long f(double long x)
{
return (1/(pow(x,3)-5));
}
main()
{
int option;

```

```

float a,b,h,n,I;
float sum=0;
cout<<"Press 1 for Simpson's and Press 2 for Trapezoidal rule:\t";
cin>>option;
cout<<"enter lower limit:\n";
cin>>a; //1
cout<<"enter upper limit:\n";
cin>>b; //6
cout<<"enter number of intervals:\n";
cin>>n; //n=6 given
h=(b-a)/n;
sum=f(b)+f(a);
switch(option)
{
case 1:
cout<<"integration by Simpson's Rule:\n";
for (int i=1;i<n;i++)
{
if (i%2==0)
{
sum=sum+2*f(a+(i*h));
}
else
{
sum=sum+4*f(a+(i*h));
}
}
}
I=(h/3)*sum;
cout<<"Definite Integration:\t"<<I<<endl;
break;
case 2:
cout<<"integration by Trapezoidal Rule:\n";
for (int i=1;i<n;i++)

```



```

        {
            sum=sum+2*f(a+(i*h));
        }
I=(h/2)*sum;
cout<<"Definite Integration:\t"<<I<<endl;
break;
default:
    {
        cout<<" error :Invalid option"<<endl;
        break;
    }
}
return 0;
}

```

Q#4

Suppose A & B be 3x3 matrices. Write C++ program which reads in entries of A & B and Prints out the entries of matrix which is

- (i) $7A-3B$ and (ii) $C=2A^t$

Program:

```

#include<iostream>
#include<math.h>
using namespace std;
main()
{
    float A[3][3],B[3][3],C[3][3],D[3][3];
    int i,j;
    cout<<"enter the values in matrix A:\n";
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            cin>>A[i][j];
        }
    }
}

```

```

}

cout<<"enter the values in matrix B:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cin>>B[i][j];
    }
}

cout<<"entered values in matrix A:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<A[i][j]<<"\t";
    }
    cout<<endl;
}

cout<<"entered values in matrix B:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<B[i][j]<<"\t";
    }
    cout<<endl;
}

for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        D[i][j]=7*A[i][j]-3*B[i][j];
        C[i][j]=A[j][i];
    }
}

```

```

}
}
cout<<" Result of 7A-3B is:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<D[i][j]<<"\t";
    }
    cout<<endl;
}
cout<<" Sum of C=2At is:\n";
for(i=0;i<3;i++)
{
    for(j=0;j<3;j++)
    {
        cout<<2*C[i][j]<<"\t";
    }
    cout<<endl;
}
return 0;
}

```

Q#5 (a)

Write C++ Program for the forced Harmonic Motion (F.H.M) of a mass attached with a spring using Euler's method under the following conditions:

($g=9.8 \text{ ms}^{-2}$, initial position=zero ,and velocity= 15 ms^{-1} time step 0.1 sec and maximum time 15 sec. $k=1 \text{ N/m}$, $m=1 \text{ Kg}$, damping coefficient= 0.5 N/ms , $w=0.01 \text{ s}^{-1}$ and $f_0=1.5 \text{ N}$).

Calculate and Print values for time , Position and velocity and aceleration. How u can change the same program for Simple harmonic motion,

Program:

```
#include<iostream>
```

```
#include<conio.h>
```

```

#include<iomanip>
#include<math.h>
using namespace std;

int main()
{
float x=0,t=0,a,tmax=15,v=15,k=1,m=1,h=0.1,b=0.5,f=1.5,w=0.01;
a=((-k*x)+(-b*v)+(f*cos(w*t)))/m;
int p=0;
float X[1000],T[1000],V[1000],A[1000];
while(t<=tmax)
{
a=((-k*x)+(-b*v)+(f*cos(w*t)))/m;
T[p]=t;
V[p]=v;
X[p]=x;
A[p]=a;
x=x+(v*h);
v=v+(a*h);
t=t+h;
p++;
}
cout<<setw(16)<<" Time "<<setw(16)<<" position "<<setw(16)<<" velocity "<<setw(16)<<" acceleration
"<<endl;
for( int i=0; i<p; i++ )
{
cout<<setw(16)<<T[i]<<setw(16)<<X[i]<<setw(16)<<V[i]<<setw(16)<<A[i]<<endl;
}
return 0;
}

```

Q#5 (b)

Write C++ Program which reads in 10 numbers into an array and display those numbers in reverse order.

Program:

```
#include <iostream>
using namespace std;
int main()
{
int arr[10];
int hold;
for(int i=0; i<10; i++)
{
cin>>arr[i];
}
for(int i=0; i<10; i++)
{
for(int j=i+1; j<10; j++)
{
if(arr[j] > arr[i])
{
hold = arr[i];
arr[i] = arr[j];
arr[j] = hold;
}
}
}
cout<<"Elements of array is in reverse order:\n";
for(int i=0; i<10; i++)
{
cout<<arr[i]<<endl;
}
return 0;}
```

All Program End (2013-2019)

Typed By: Mha Physics

Institution: Govt. Post Graduate College Sheikhpura.

E-Mail: mhaphysics999@gmail.com

Blogger: <http://mhaphysics.blogspot.com>

You Tube Channel Name: [Mha Physics](#)

Facebook page & group: [Mha Physics](#)

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Remember In Your Prayers.